Research Article

EXPLAINABLE ANOMALY DETECTION IN INFORMATION SYSTEMS STATE OF THE ART AND APPLICATION TO TIMESHEET AND PAYROLL SYSTEMS

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ABSTRACT

Timesheet and payroll information systems play a critical role within organizations, as they handle sensitive data that directly affect employees, regulatory compliance, and financial governance. Ensuring the reliability and integrity of such data therefore represents a major managerial challenge.
However, these systems are exposed to various anomalies arising from data entry errors, inconsistencies, or fraudulent behavior, which may compromise data quality and undermine user trust. Traditional control mechanisms increasingly reach their limits in the face of growing data volume and system complexity.
In response to these challenges, recent literature has increasingly focused on automated and explainable anomaly detection approaches. In sensitive domains such as payroll management, anomaly detection alone is insufficient and must be complemented by robust explainability mechanisms.
This study provides a structured state-of-the-art review of explainable anomaly detection approaches applied to timesheet and payroll information systems.

Introduction

Information systems are fundamental to organizational management processes, particularly timesheet and payroll systems, which process sensitive data related to working hours and employee remuneration. These systems require a high level of reliability to ensure regulatory compliance and sound financial governance [1, 2, 3]. Nevertheless, they may be affected by anomalies resulting from errors, inconsistencies, or abnormal behaviors, potentially leading to operational disruptions such as payment delays. In this context, traditional control mechanisms—typically based on predefined rules and manual verification—are increasingly inadequate given the growing complexity of data and information systems [2], [3]. Automatic anomaly detection therefore emerges as a promising approach to strengthen data control mechanisms in management systems. However, in sensitive domains such as payroll, the mere identification of anomalies is insufficient. Users must be able to understand the underlying causes of alerts in order to make informed and defensible decisions. Consequently, anomaly explainability has become a central concern for enhancing user trust and ensuring the auditability of automated detection systems [4], [5].
The objective of this bibliographic study is to provide a structured state-of-the-art overview of explainable anomaly detection approaches in information systems and to assess their relevance for timesheet and payroll applications.

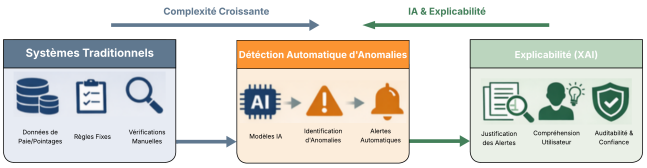

Figure 1: Transition from traditional controls to explainable anomaly detection

1 Anomaly detection: concepts and challenges

1.1 Definition of anomalies

In management information systems, an anomaly can be defined as a deviation from expected operational behavior, as inferred from historical data, business rules, or organizational processes. Notably, the definition of normality remains context-dependent [6], [7].
It is important to distinguish anomalies from related concepts such as errors, which are generally unintentional and linked to data entry or processing issues [8]; inconsistencies, which arise from contradictions between information or business rules [9]; and fraud, which involves intentional rule violations for personal or organizational gain [2], [10].
An anomaly thus represents a generic warning signal that does not presuppose its cause or intent but aims to draw the attention of human actors to situations requiring in-depth investigation. This justifies the need for reliable and interpretable detection mechanisms [11].

1.2 Typologies of anomalies

The literature commonly identifies three main categories of anomalies in management systems. Statistical anomalies correspond to atypical numerical values relative to reference behavior and are frequently observed in transactional or temporal datasets [8], [10]. Behavioral anomalies relate to unusual patterns of actions or user profiles, which can be detected through temporal analysis or comparisons among similar entities [12], [2]. Finally, logical anomalies arise from violations of business rules or constraints, even when individual data values appear consistent [7], [9].
This typology highlights that anomaly detection relies on the analysis of data, behaviors, and business rules, particularly within ERP environments as well as timesheet and payroll systems [7], [6].

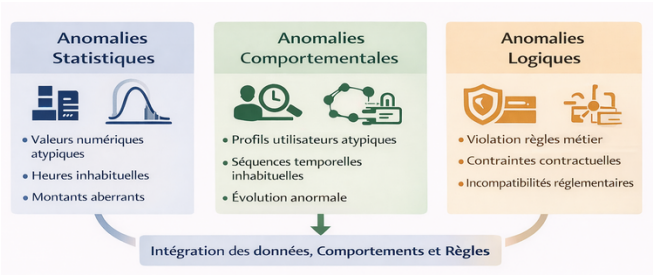

Figure 2: Typology of anomalies in ERP systems

2 Anomaly detection approaches in management systems

Scientific literature distinguishes several families of anomaly detection approaches in management systems, which differ in terms of automation level, adaptability, and interpretability. These approaches are generally categorized into rule-based methods, statistical and exploratory methods, and advanced approaches applied to ERP environments. The choice among them largely depends on the explainability and auditability requirements of the application context [13].

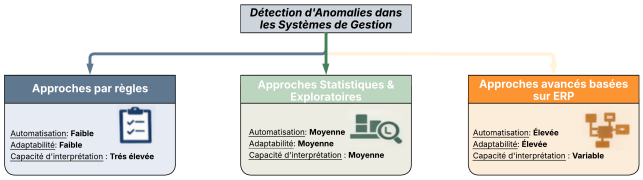

Figure 3: Anomaly detection approaches in information systems

2.1 Rule-based approaches

Rule-based approaches represent the earliest form of control in management information systems. They rely on the explicit formalization of business rules derived from regulatory or organizational frameworks, thereby defining the normal operating conditions of administrative and financial processes [1].
In timesheet and payroll systems, such rules enable the verification of declared working hours and the compliance of remuneration calculations [2], [3]. Their main strengths lie in their simplicity, transparency, and ease of auditing, as rules are directly understandable by business stakeholders [4].
However, these approaches are poorly suited to evolving behavioral patterns, difficult to maintain at scale, and often ineffective in detecting novel or subtle anomalies. In complex and dynamic environments, they may also generate a high number of false positives [10], [13], also motivating the development of more flexible, data-driven methods.

2.2 Statistical and exploratory approaches

Statistical approaches detect anomalies by identifying atypical values relative to a reference behavior through data distribution analysis. They are particularly well suited to the tabular and numerical data commonly found in management systems, including timesheet and payroll datasets [6].
These methods facilitate the identification of significant deviations in variables such as working hours or compensation amounts [8]. Profile-based analysis extends this perspective by comparing the behavior of an entity to that of a reference group in order to detect persistent anomalies [12].
Nevertheless, these approaches often provide limited explanatory power, typically indicating only that a value is atypical. Some studies propose complementing them with post-hoc explanation mechanisms to clarify the contribution of specific variables to the detected anomaly [14]. Despite these advances, their reliance on predefined thresholds and context-dependent definitions of normality limits their robustness and interpretability in sensitive domains such as payroll management [10].

2.3 Advanced approaches applied to ERP

In ERP systems, anomaly detection increasingly relies on the analysis of transactional logs and business process execution traces [7]. These approaches model event sequences, module dependencies, and user behavior patterns in order to detect process-level anomalies rather than simple statistical outliers. Log-based ERP analysis enables the identification of atypical transaction sequences, violations of process flows, or inconsistent user behaviors over time [2], [12]. However, the complexity and heterogeneity of ERP data make anomaly interpretation challenging, thereby requiring explainability mechanisms capable of linking detected anomalies to specific process steps or violated business rules [15].

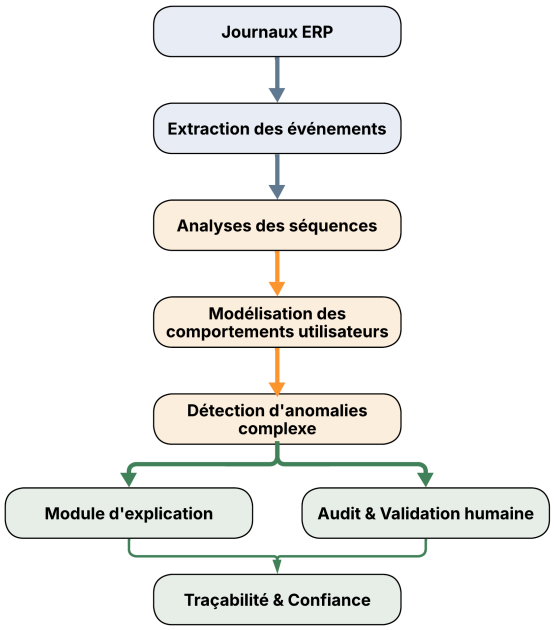

Figure 4: Anomaly detection based on ERP log analysis

These approaches offer improved detection capabilities but simultaneously reinforce the need for explainable tools, particularly in payroll contexts where each anomaly must be precisely justified and traceable [11], [7], [3].

	Rule-based	Statistical	Advanced ERP
Automation	Low	Medium	High
Adaptability	Low	Medium	High
New anomaly detection	Low	Medium	High
Interpretability	Very high	Medium	Variable
Complexity	Low	Medium	High
Auditability	Strong	Medium	Strong if explainable

Table 1: Comparison of anomaly detection approaches

3 Explainability of anomalies

3.1 Why explain an anomaly?

In critical management systems, an unexplained alert is of limited practical value. Explainability transforms automated detection into actionable insight by identifying the factors that led to the anomaly and thereby facilitating human analysis [4], [5].
Explainability also contributes to ensuring the traceability and auditability of automated decisions, which is essential in sensitive domains such as payroll and timesheet management [4]. Furthermore, it enhances user acceptance of automated detection systems and supports the integration of human validation into decision-making processes [16].

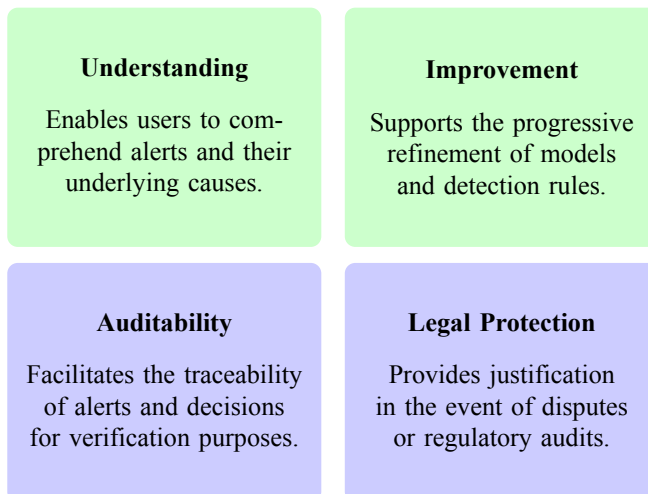

Figure 5: Benefits of explainability

3.2 Types of explanations proposed in literature

The literature identifies several families of explainability methods applied to anomaly detection, including post-hoc explanation techniques such as LIME and SHAP, as well as intrinsically interpretable models.
LIME relies on local model approximation by constructing an interpretable surrogate model around the anomalous observation, enabling the identification of the most influential variables in a local context. SHAP, based on Shapley value theory, assigns each variable an additive contribution to the model's decision, thereby providing a consistent quantitative measure of attribute influence on anomaly detection [4], [6], [11].
In parallel, intrinsically explainable models—such as rule-based systems, logical constraints, or hybrid approaches combining symbolic reasoning and machine learning—integrate interpretability directly into the detection process. These models explicitly link anomalies to violations of business rules or constraints, making them particularly suitable for ERP and payroll systems subject to stringent regulatory requirements [7], [9].

The literature also distinguishes between global explanations, which describe the overall behavior of a model, and local explanations, which focus on a specific alert. Local explanations are generally preferred in timesheet and payroll systems, where each anomaly must be individually analyzed and validated [5], [6].

3.3 Contributions and limits of explainable approaches

Explainable approaches enhance the understanding of alerts, facilitate human validation, and strengthen the traceability of automated decisions, thereby promoting their integration into audit and control processes [5], [11], [3].
However, the quality and usefulness of explanations largely depend on the underlying models and the application context. A persistent trade-off exists between detection performance and interpretability, and the objective evaluation of explanation quality remains an open research challenge [5], [6]. These limitations confirm that explainability continues to represent a central research focus in management information systems.

4 Application to timesheet and payroll systems

Timesheet and payroll systems rely on highly structured, tabular, and temporal data derived from HR and ERP environments, combining individual, contractual, and regulatory information [1], [2]. The strong dependence on business rules makes the definition of normality highly contextual, while any anomaly may entail financial, social, or legal consequences.
Within these systems, anomalies may be statistical—for example, an unusually high number of recorded working hours—or behavioral, such as the recurrent use of overtime during specific periods [12]. Logical anomalies are particularly critical in payroll contexts and typically result from explicit violations of business rules, such as inconsistencies between overtime and employment contracts, unauthorized bonus allocations, or discrepancies between recorded attendance and declared absences [7], [9].
Reliability and compliance requirements therefore mandate that such anomalies be detected prior to the final validation of payroll processes [3], [7]. In this context, detection approaches are only fully effective when complemented by explanation mechanisms that enable the justification of alerts to both business stakeholders and regulatory authorities [4], [5].
Explainable approaches support the analysis and validation of alerts by linking anomalies to business rules, specific variables, or behavioral deviations, thereby reducing investigation time [7], [8]. Local explanations are particularly relevant in payroll management, where each alert must be individually examined [6], [11]. The literature also emphasizes the importance of human validation to mitigate false positives and enhance the acceptability of automated systems [8], [12], [16].
Nevertheless, research specifically focused on HR systems remains limited. Many existing approaches are adapted from other ERP contexts without fully accounting for the specific characteristics of payroll data, notably their mixed, temporal, and highly regulated nature [2], [14], [15].

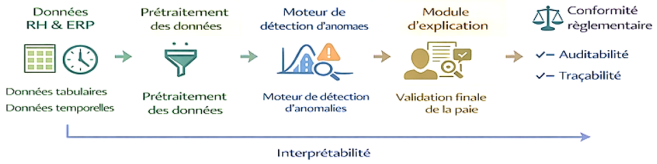

Figure 6: Explainable anomaly detection approach

5 Discussion and research perspectives

The literature indicates that automated anomaly detection constitutes a significant lever for strengthening control mechanisms in management information systems, particularly in ERP and HR environments characterized by complex and interconnected data structures. However, in sensitive contexts such as timesheet and payroll management, anomaly detection alone is insufficient and must be complemented by mechanisms that support alert interpretation and justification.
Explainability therefore emerges as a key factor in enhancing user trust, facilitating auditability, and supporting human decision-making. Nevertheless, several challenges remain, including the trade-offs between detection performance and interpretability, as well as the difficulty of objectively evaluating explanation quality.
Future research directions primarily focus on improving explainability techniques for mixed and temporal data, strengthening the integration of business rules and human validation, and developing fully auditable detection frameworks compliant with the regulatory constraints of critical management systems.

6 Conclusion

This bibliographic study presented a state-of-the-art review of explainable anomaly detection approaches in information systems, with particular emphasis on timesheet and payroll applications. The literature shows that traditional rule-based and statistical approaches, while still useful, are increasingly limited in addressing the growing complexity of data and management processes.
Advanced approaches applied to ERP environments offer enhanced detection capabilities but simultaneously reinforce the need for explainability to ensure understanding, user trust, and decision auditability, especially in sensitive contexts such as payroll management. Finally, future research perspectives highlight the importance of adapting explainability mechanisms to mixed and temporal data, strengthening the integration of business rules and human validation, and developing fully compliant anomaly detection frameworks.

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